12.19 Random Points in the Plane

WORKSHEET

The question

STEP II 2012, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s2-q13

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself A Poisson random variable with mean μ takes the value 0 with what probability?

STAGE 1 No supermarket nearby

Covers: no supermarkets in a circle

Ask yourself The number of supermarkets in a circle of radius y is Poisson. What is its mean? What is the probability that it is 0?

STAGE 2 The distribution of the nearest distance

Covers: the distribution of Y

Pause and try $Y \geq y$ means there is no supermarket within distance y. So what is $\mathrm{P}(Y < y)$? Differentiate to get the density.

STAGE 3 The mean

Covers: the mean

Ask yourself Write $\mathrm{E}(Y)$ as an integral. Substitute so that the exponent becomes $-\frac{1}{2}x^2$, and use the integral given in the question.

STAGE 4 The variance

Covers: the variance

Ask yourself For $\mathrm{E}(Y^2)$ the power of y in the integrand is odd, so substitute $u = \pi k y^2$. Then use $\mathrm{Var}(Y) = \mathrm{E}(Y^2) - \big(\mathrm{E}(Y)\big)^2$.

What's really going on

The Poisson process in the plane

A spatial Poisson process with intensity k is a random scatter of points such that the numbers in disjoint regions are independent, and the number in a region of area A is Poisson with mean kA. It is what you get from dropping a huge number of points independently and uniformly over a huge area, with the right average density: the "law of rare events" that also explains derangements (Section 9.8). It is the natural model of complete spatial randomness, and real patterns are judged against it. The *Clark–Evans test*, devised by two ecologists in 1954, compares the observed mean nearest-neighbour distance with $\frac{1}{2\sqrt{k}}$: much smaller suggests clustering (seeds falling near their parent tree), much larger suggests regular spacing (territorial birds, or competition between plants for water).

In three dimensions

The same calculation in three dimensions gives $\mathrm{P}(Y > y) = e^{-\frac{4}{3}\pi k y^3}$ for the distance to the nearest star, molecule or galaxy. Subrahmanyan Chandrasekhar used it in 1943 to study the gravitational forces on a star from its random neighbours, and physicists use it for the mean free path of molecules in a gas.

Rayleigh

The density $\frac{y}{\sigma^2} e^{-y^2/(2\sigma^2)}$ is named after Lord Rayleigh, who met it in 1880 in the theory of sound: it describes the amplitude of the sum of many waves with random phases. It also gives the distance from the origin of a point whose two coordinates are independent normal random variables, which is another way to see why it appears here (exercise 3). Today it models the fading of mobile-phone signals, which reach the

receiver along many paths with random phases.

Try this next

1. **The median.** Find the median distance to the nearest supermarket, and compare it with the mean.
2. **The second nearest.** Find the distribution of the distance to the *second* nearest supermarket. (*Hint:* $\mathrm{P}(Y_2 > y)$ is the probability of at most one supermarket within y.)
3. **Normal coordinates.** If U and V are independent $\mathrm{N}(0, \sigma^2)$, show that $\sqrt{U^2 + V^2}$ has density $\frac{r}{\sigma^2} e^{-r^2/(2\sigma^2)}$. Which value of σ gives the supermarket distribution?
4. **Clark–Evans.** In a survey of a 1 km^2 wood containing 400 trees, the mean distance from a tree to its nearest neighbour is 35 m. Is this consistent with complete spatial randomness, or does it suggest clustering or spacing?

Further reading

- **Peter Diggle, *Statistical Analysis of Spatial Point Patterns*** (Arnold). The standard reference, with the Clark–Evans test and its successors.
- **John Kingman, *Poisson Processes*** (Oxford University Press). A beautiful short book on Poisson processes in any number of dimensions.
- **Geoffrey Grimmett & David Stirzaker, *Probability and Random Processes*** (Oxford University Press), the chapter on the Poisson process.

The full guided solution is in Section 12.19:
https://beyondstep.keithmcnulty.org/ch12/spatial-poisson.html